ENGR 095

Fall 2026 - Jenn Winikus & Nick Strandwitz

Diane Ngo

8 October 2026

Sensing the World - Project Reflection 3

In the final part of the “Sensing the World” module, we were tasked as groups to choose a humanitarian issue and build and design a prototype that would help our chosen issues. My group decided to build a flood gate that detected water and rising humidity to automatically close and protect vulnerable communities that suffer flash floods. It would make it so that no human would have to be present to close the flood gate themself, allowing for more safety to the people. First, we planned out our project on a worksheet, describing our goals, how we would build it, responsibilities, and more. Then, we spent 3-4 days in the lab building our prototypes. My group decided to split things up, Diego did all of the programming and most of the electrical part while me and Artur worked on the mechanical building of the project. Artur and I first used cardboard to build our base and plan out the structure of our design as Diego worked on the wiring of the bread board. As a group, we planned out what electronic parts we would use and where they'd go as Diego made files to test each part individually. At the end, we put it all together and tested the programming and build together and were successful after a few tweaks.

Throughout the project, we learnt a lot about the design process and engineering lessons., First, we had issues with our servo motor and had to switch to the stepper motor as the servo motor only went up to 180 degrees and we needed a lot more than 180 degrees to pull up the gate. Transitioning to the stepper motor meant we had to make a pulley and we learned a lot about pulley tension and we needed a counter weight as well as another wheel for the string/pulley so it wouldn't get caught on the cardboard structure. We also had issues with the

structure not being as stabler so we used a lot of more tape and cardboard to reinforce it so our builds wouldn't wobble as much as it was taller and bigger. Making our gate, we realized we needed guide rails so that the gate would just be loose and added those, but then because of the pulley, the string pulled it up but it still caused it to lean on one side so we had to add more counter weight to the other side of the gate. We, like most groups, also had some issues with our humidity PEDOT sensor. However, we realized that when blowing on it, we blew way too hard or too close and that was why it caused the gate to stay closed for a long time. We were able to change the threshold values and fixed the issue. In the end, we managed to get through all of the issues we encountered with teamwork and persistence and have a reliable successful prototype of our flood gate.